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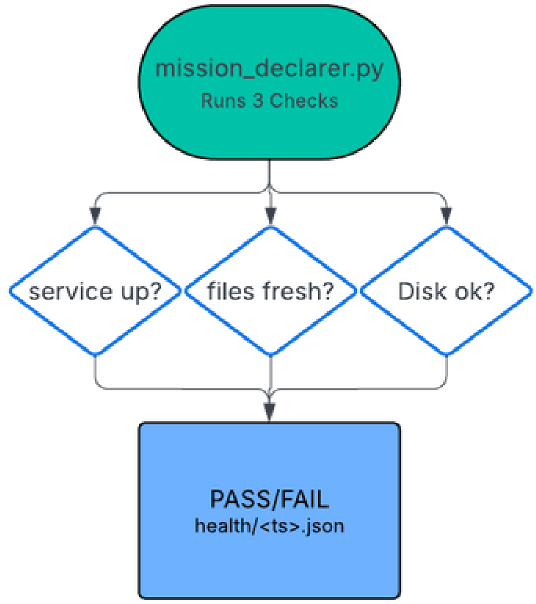
VIASAT - SATELLITE CYBER ATTACK & DEFENSE (SAT-CYAD)



What?

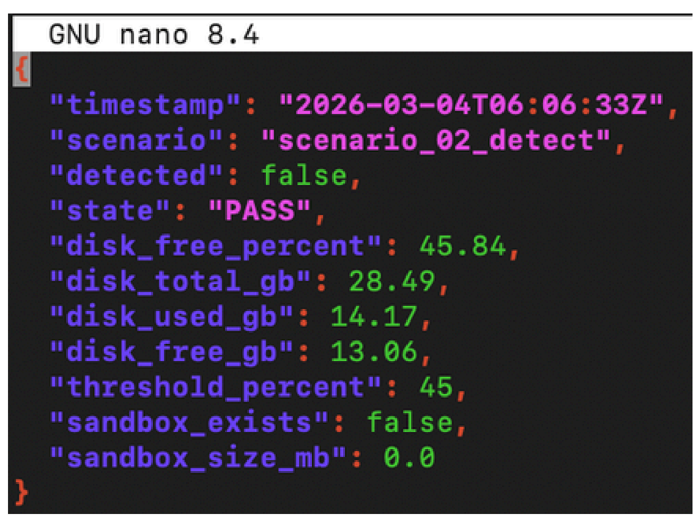
- Built reliability infrastructure for a live Geostationary Operational Environmental Satellite (GOES) weather satellite ground station
- Designed fault simulation system to test pipeline resilience against disruption scenarios
- Created run-books for smooth operater use on ground station

Results

- Demonstrated pipeline disruption and disk pressure fault scenarios with automated recovery verification
- 2 fault scenarios automated, 0 manual intervention required

How?

- Automated service management with systemd (boot startup, crash recovery, dependency ordering)
- Built 3-layer fault simulation: health monitoring, fault injection, and orchestration
- Used Python + Bash to produce structured JSON evidence artifacts per scenario



GNU nano 8.4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{
  "timestamp": "2026-03-04T06:06:33Z",
  "scenario": "scenario_02_detect",
  "detected": false,
  "state": "PASS",
  "disk_free_percent": 45.84,
  "disk_total_gb": 28.49,
  "disk_used_gb": 14.17,
  "disk_free_gb": 13.06,
  "threshold_percent": 45,
  "sandbox_exists": false,
  "sandbox_size_mb": 0.0
}
```